

Two counterexamples in orthogonality of measures and states

Candidate full-counterexample packet

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Status. Candidate full counterexamples; likely valid; human review recommended.

1 Source questions

Mejak asks whether, for a separable unital C^* -algebra A , the common strong-orthogonal complement of every analytic strongly orthogonal family of states is comeagre [1]. The exact question (PDF page 18) is below.

3.2. Comeagreness of witnesses to non-maximality for states. Let A be a separable unital C^* -algebra. As we have seen above, there is a metric d on $S(A)$, defined by Bures in [Bur69], which satisfies that for $\phi, \psi \in S(A)$ it holds that $\phi \perp \psi$ if and only if $d(\phi, \psi) = \sqrt{2}$.

This gives us hope that Theorem 3.1 might apply to $\frac{1}{\sqrt{2}}d$. However, we do not know whether items (2) and (3) of Theorem 3.1 are satisfied. Even if this approach fails, it might still be the case that the following has a positive answer.

Question 3.5. *Is it the case for a separable unital C^* -algebra A (or for a class of C^* -algebras satisfying some additional properties), that for any analytic strongly orthogonal family $\mathcal{A} \subseteq S(A)$, the set*

$$\mathcal{A}^\perp := \{\psi \in S(A) : (\forall \phi \in \mathcal{A}) \psi \perp \phi\}$$

is comeagre?

3.3. Measure on the space of measures. For this subsection we work on the Cantor

The universal statement has a negative answer, already for $M_2(\mathbb{C})$ and also for an infinite-dimensional separable unital algebra.

The source next defines a probability measure Λ on $P(2^{\mathbb{N}})$ by putting independent uniform splitting parameters at the vertices of the binary tree. Question 3.6 asks whether $\Lambda(\mathcal{A}^\perp) > 0$ for every analytic orthogonal family. Claim 3.7 then asserts $\Lambda(\mu^\perp) < 1$ for every fixed μ . The source passage (PDF page 19) is reproduced below.

Question 3.6. Suppose that $\mathcal{A} \subseteq P(2^{\mathbb{N}})$ is an analytic orthogonal family. Does it hold that $\Lambda(\mathcal{A}^{\perp}) > 0$?

One should not get one's hopes too high and wonder whether it could be that $\Lambda(\mathcal{A}^{\perp}) = 1$, as this turns out to be false.

Claim 3.7. For any $\mu \in P(2^{\mathbb{N}})$, $\Lambda(\mu^{\perp}) < 1$.

Sketch of proof. Fix any $\mu \in P(2^{\mathbb{N}})$ and suppose for contradiction that $\Lambda(\mu^{\perp}) = 1$. Fix also some arbitrary small $\varepsilon \in (0, 1/2)$. For $s \in 2^{<\mathbb{N}}$, let $U_s := \{x \in 2^{\mathbb{N}} : s \subseteq x\}$. By repeated use of Fubini's theorem and the fact that for $B \subseteq [0, 1]$, $\lambda(B) = 1$ implies that B is dense in $[0, 1]$, we get that there is some $\nu \in \mu^{\perp}$ so that for all $s \in 2^{<\mathbb{N}}$ it holds that

$$\nu(U_s) \in ((1 - \varepsilon)\mu(U_s), (1 + \varepsilon)\mu(U_s)).$$

Since open sets are disjoint unions of basic open sets, the same holds for all open $U \subseteq 2^{\mathbb{N}}$. But then ν and μ are not orthogonal, which is a contradiction. $\square$

Claim 3.7 is false: the value is always exactly one. This also gives a full and stronger answer to Question 3.6 for every countable family.

2 Failure of state-space comeagreness

For a state ϕ on A , its support is the support projection of its normal extension to A^{**} . The source defines strong orthogonality by $\phi \perp \psi$ if and only if $(\text{supp } \phi)(\text{supp } \psi) = 0$.

Proposition 1 (full-support obstruction). *Let A be a nonzero separable unital C^* -algebra. If A has a state ϕ with $\text{supp } \phi = 1$ in A^{**} , then the universal assertion in Question 3.5 is false. In fact, the closed analytic strongly orthogonal family $\mathcal{A} = \{\phi\}$ satisfies*

$$\mathcal{A}^{\perp} = \emptyset.$$

Proof. A singleton is strongly orthogonal vacuously and is closed in the compact metrizable state space. Every state ψ has nonzero support, since $\psi(1) = 1$. Hence

$$(\text{supp } \phi)(\text{supp } \psi) = 1 \cdot \text{supp } \psi = \text{supp } \psi \neq 0.$$

No state is strongly orthogonal to ϕ . The empty set is not comeagre in the nonempty Baire space $S(A)$. $\square$

Corollary 1. *Question 3.5 has a negative answer for $A = M_2(\mathbb{C})$.*

Proof. The normalized trace has density matrix $I/2$ and therefore support I . Apply Proposition 1. $\square$

The obstruction is not confined to finite-dimensional state spaces.

Corollary 2 (infinite-dimensional counterexample). *Question 3.5 also has a negative answer for the separable infinite-dimensional unital algebra*

$$A = \mathcal{K}(\ell_2) + \mathbb{C}I.$$

Proof. The bidual identifies with $B(\ell_2) \oplus \mathbb{C}$, under the embedding

$$k + \lambda I \longmapsto (k + \lambda I, \lambda).$$

Let ρ be the diagonal trace-class operator with diagonal $2^{-1}, 2^{-2}, \dots$. The normal state

$$\Phi(T, z) = \frac{1}{2} \operatorname{Tr}(\rho T) + \frac{1}{2} z \quad (T \in B(\ell_2), z \in \mathbb{C})$$

has support $(I, 1)$: ρ has trivial kernel and both direct-summand coefficients are positive. Its restriction to A therefore has support one in A^{**} . Proposition 1 applies. $\square$

This also settles the source's immediately preceding uncertainty in the negative for general algebras: item (3) of its abstract theorem, applied to the normalized Bures metric, fails for the normalized trace on M_2 , whose strong-orthogonal complement is empty.

3 The random splitting measure is singular to every fixed measure

For a finite binary word s , let U_s be its cylinder in $2^{\mathbb{N}}$. Under the probability law used to define Λ , choose independent $V_s \sim \operatorname{Unif}[0, 1]$ and recursively set

$$\nu(U_{s \smallfrown 0}) = \nu(U_s) V_s, \quad \nu(U_{s \smallfrown 1}) = \nu(U_s)(1 - V_s).$$

Theorem 1 (reverse of Claim 3.7). *For every fixed $\mu \in P(2^{\mathbb{N}})$,*

$$\Lambda(\mu^\perp) = 1.$$

Consequently, if $\mathcal{A} \subseteq P(2^{\mathbb{N}})$ is countable, then

$$\Lambda(\mathcal{A}^\perp) = 1,$$

without any orthogonality or definability assumption on $\mathcal{A}$.

Proof. For two probability measures μ, ν , let

$$H(\mu, \nu) = \int \sqrt{\frac{d\mu}{d\xi} \frac{d\nu}{d\xi}} d\xi, \quad \xi = \mu + \nu,$$

be their Hellinger affinity. It is zero exactly when $\mu \perp \nu$. For the level- n cylinder partition put

$$H_n(\mu, \nu) = \sum_{|s|=n} \sqrt{\mu(U_s) \nu(U_s)}.$$

Cauchy–Schwarz on each cylinder gives $H(\mu, \nu) \leq H_n(\mu, \nu)$.

Fix s of length n . The mass $\nu(U_s)$ is a product of n independent factors, each distributed uniformly on $[0, 1]$; using $1 - V_t$ instead of V_t does not change the distribution. Therefore

$$\mathbb{E} \sqrt{\nu(U_s)} = \left(\int_0^1 \sqrt{u} du \right)^n = \left(\frac{2}{3} \right)^n.$$

It follows that

$$\begin{aligned} \mathbb{E} H_n(\mu, \nu) &= \left(\frac{2}{3} \right)^n \sum_{|s|=n} \sqrt{\mu(U_s)} \\ &\leq \left(\frac{2}{3} \right)^n \sqrt{2^n \sum_{|s|=n} \mu(U_s)} = \left(\frac{2\sqrt{2}}{3} \right)^n. \end{aligned}$$

Since $2\sqrt{2}/3 < 1$ and $0 \leq H(\mu, \nu) \leq H_n(\mu, \nu)$ for every n , we obtain $\mathbb{E} H(\mu, \nu) = 0$. Thus $H(\mu, \nu) = 0$ almost surely, proving the first assertion. A countable intersection of these probability-one events proves the second. $\square$

The equality-one phenomenon also occurs simultaneously for a natural uncountable analytic orthogonal family.

Corollary 3 (all Dirac masses simultaneously). *Let $\mathcal{D} = \{\delta_x : x \in 2^{\mathbb{N}}\}$. Then $\mathcal{D}$ is an uncountable compact orthogonal family and*

$$\Lambda(\mathcal{D}^\perp) = 1.$$

Proof. For a random splitting measure ν , put $S_n = \sum_{|s|=n} \nu(U_s)^2$. Conditional on level n , each mass m splits into mV and $m(1 - V)$, and

$$\mathbb{E}[V^2 + (1 - V)^2] = \frac{2}{3}.$$

Hence $\mathbb{E}S_n = (2/3)^n$. If ν had an atom of mass at least ε , then $S_n \geq \varepsilon^2$ for every n . Markov's inequality followed by $n \rightarrow \infty$ shows that this event has probability zero. Taking the union over $\varepsilon = 1/m$ proves that ν is atomless almost surely. An atomless measure is singular to every Dirac mass simultaneously. $\square$

4 Error location, scope, and literature boundary

The sketch of Claim 3.7 tries to combine the full-measure hypothesis with infinitely many cylinder constraints. Full measure gives density in each finite coordinate cylinder, but it does not force intersection with the measure-zero tail set on which all multiplicative constraints hold. For a Dirac target, those constraints would force positive mass on its supporting point, while Theorem 1 shows that the random measure is almost surely singular to it. The Hellinger calculation identifies the failure without relying on this diagnosis.

Theorem 1 does not settle Question 3.6 for an arbitrary uncountable analytic orthogonal family. Fixed-measure null sets need not have a common full-measure intersection over uncountably many parameters. The non-orthogonality relation has countable vertical sections on a pairwise singular family, but null horizontal sections and countable vertical sections alone do not make its projection null; the diagonal relation is the basic warning. The state result disproves the broad universal wording of Question 3.5, while the search for useful positive classes remains meaningful.

The latest arXiv version is v5, dated 13 May 2022. Bounded exact-phrase searches on 11 August 2026 found Claim 3.7 repeated in the author's 2026 PhD thesis but found no correction, no prior reverse Hellinger theorem for this Λ , and no recorded full-support-state counterexample to Question 3.5. Novelty confidence is moderate.

Human-review recommendation. Check the convention that singleton families count as strongly orthogonal, the standard bidual identification for $\mathcal{K}(\ell_2) + CI$, and the single constant $2\sqrt{2}/3 < 1$ in the Hellinger calculation. The latter is the core of the correction to Claim 3.7.

References

- [1] S. Mejak, *Orthogonality of measures and states*, arXiv:2204.02767v5, 2022.